Module 2: Superconducting Low-Temperature Physics

BCS gap physics, quasiparticles, recombination, diffusion, and trapping

Physics note for FEM_BallisticDiffusion_PINN

Purpose. This note explains how the transport model changes when the device is cooled from room temperature to superconducting-qubit operating temperature. The room-temperature heat-carrier picture remains useful, but superconductivity introduces a new energy scale, the superconducting energy gap, and a new failure-relevant carrier population: quasiparticles. The practical goal is to obtain reduced equations suitable for FEM simulation of quasiparticle poisoning and trap/heat-sink design.

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Symbols and acronyms used in this document

Symbol / acronym	Name	Meaning in this document
BCS	Bardeen-Cooper-Schrieffer theory	Microscopic theory of conventional superconductivity based on Cooper pairing.
T_c	Critical temperature	Temperature below which a material becomes superconducting.
T	Temperature	Local or bath temperature.
$\Delta(T)$	Superconducting energy gap	Minimum quasiparticle excitation energy scale in a conventional superconductor.
Δ_0	Zero-temperature gap	Approximate value of $\Delta(T)$ as $T \to 0$.
N_0	Normal-state density of states	Single-spin electronic density of states at the Fermi level, depending on convention.
n_{qp}	Quasiparticle density	Density of broken-pair excitations in the superconducting film.
$n_{\mathrm{qp}}^{\mathrm{eq}}$	Equilibrium quasiparticle density	Thermal-equilibrium quasiparticle density predicted by the gap and temperature.
D_{qp}	Quasiparticle diffusion coefficient	Effective spatial diffusion coefficient for quasiparticles.
R	Recombination coefficient	Coefficient for two-quasiparticle recombination into a Cooper pair.
Γ_{trap}	Trap removal rate	Position-dependent quasiparticle removal rate due to normal-metal traps or engineered sinks.
S_{qp}	Quasiparticle source	Generation rate density from photons, phonons, heating, or other nonequilibrium events.
C_s	Superconducting heat capacity	Electronic heat capacity in the superconducting state.
k_s	Superconducting thermal conductivity	Effective thermal conductivity in the superconducting state.
Ω_{JJ}	Junction-sensitive region	Effective volume around the Josephson junction where quasiparticles are most harmful.

Symbol / acronym	Name	Meaning in this document
JJ	Josephson junction	Weak link between superconductors that provides nonlinearity to a superconducting qubit.
QP	Quasiparticle	Excitation formed when Cooper-pair condensate order is locally broken or excited.

1 Why cooling changes the physics

At room temperature, electrons and phonons can often be modeled as heat carriers whose energy density and flux can be connected to temperature, conductivity, and mean free path. At superconducting-qubit temperatures, the electronic system is no longer a normal metal. A macroscopic condensate of Cooper pairs forms, and single-particle excitations require energy at least on the order of the superconducting energy gap Δ.

This changes the modeling target. The design question is not only how to move heat away. It is also how to reduce the density of nonequilibrium quasiparticles near the Josephson junction. A small number of quasiparticles in the wrong place can matter because they can tunnel across the junction, change parity, introduce loss, and disrupt qubit operation.

2 Minimal BCS energy scale

For a conventional weak-coupling BCS superconductor, the zero-temperature gap is approximately

$$\Delta_0 \approx 1.764\, k_{\mathrm{B}} T_c. \tag{2.1}$$

A commonly used interpolation for the temperature dependence is

$$\Delta(T) \approx \Delta_0 \tanh\left[1.74\sqrt{\frac{T_c}{T} - 1}\right], \qquad 0 < T < T_c. \tag{2.2}$$

This formula is not the full self-consistent BCS gap equation, but it is useful for engineering-level reduced modeling.

The key physical point is exponential suppression. At temperatures well below T_c, thermal quasiparticles are rare because creating them costs energy Δ.

3 Equilibrium quasiparticle density

For $k_B T \ll \Delta$, the equilibrium quasiparticle density has the approximate asymptotic form

$$n_{qp}^{eq}(T) \approx 2N_0\sqrt{2\pi k_B T \Delta(T)}\,\exp\left[-\frac{\Delta(T)}{k_B T}\right]. \tag{3.1}$$

The exponential factor is the dominant physics. A small reduction in $\Delta/k_B T$ can strongly increase the thermal quasiparticle population.

Interpretation. In an ideal cold superconductor, the thermal quasiparticle density should be exponentially small. If experiments observe more quasiparticles than Eq. (3.1) predicts, that is evidence for nonequilibrium generation: stray photons, cosmic/radiation events, phonon bursts, substrate heating, imperfect shielding, or other environmental couplings.

4 Reduced quasiparticle transport equation

The first FEM-ready quasiparticle model is a diffusion-reaction-trapping equation:

$$\frac{\partial n_{qp}}{\partial t} = \nabla\cdot(D_{qp}\nabla n_{qp}) - R\left(n_{qp}^2 - \left(n_{qp}^{eq}\right)^2\right) - \Gamma_{trap}(\boldsymbol{r})n_{qp} + S_{qp}(\boldsymbol{r},t). \tag{4.1}$$

Each term has a distinct physical meaning:

- $\nabla\cdot(D_{qp}\nabla n_{qp})$ spreads quasiparticles spatially.
- $R(n_{qp}^2 - (n_{qp}^{eq})^2)$ drives recombination toward equilibrium. The quadratic dependence appears because recombination requires two quasiparticles.
- $\Gamma_{trap}(\boldsymbol{r})n_{qp}$ removes quasiparticles in trap regions.
- $S_{qp}(\boldsymbol{r},t)$ generates quasiparticles from deposited energy or nonequilibrium excitation.

This is the central equation for Module 2 and Module 3. It is intentionally simpler than a full kinetic equation, but it is ideal for geometry-level trap studies.

5 Where the recombination term comes from

Quasiparticle recombination is a two-body process. If n_{qp} is the density of quasiparticles, then the probability of finding two quasiparticles in the same effective interaction volume scales as n_{qp}^2.

Therefore the simplest recombination law is

$$\left.\frac{\partial n_{\mathrm{qp}}}{\partial t}\right|_{\mathrm{rec}} = -R n_{\mathrm{qp}}^2. \tag{5.1}$$

However, if the material is in thermal equilibrium at temperature T, the density should not decay to zero; it should decay to $n_{\mathrm{qp}}^{\mathrm{eq}}(T)$. A detailed-balance-compatible reduced form is therefore

$$\left.\frac{\partial n_{\mathrm{qp}}}{\partial t}\right|_{\mathrm{rec}} = -R\left(n_{\mathrm{qp}}^2 - \left(n_{\mathrm{qp}}^{\mathrm{eq}}\right)^2\right). \tag{5.2}$$

If $n_{\mathrm{qp}} > n_{\mathrm{qp}}^{\mathrm{eq}}$, the term is negative and recombination dominates. If $n_{\mathrm{qp}} < n_{\mathrm{qp}}^{\mathrm{eq}}$, the term is positive, representing thermal generation toward equilibrium.

6 Trap term and physical meaning

A normal-metal quasiparticle trap is represented by a spatially localized removal rate $\Gamma_{\mathrm{trap}}(\boldsymbol{r})$. In the simplest model,

$$\Gamma_{\mathrm{trap}}(\boldsymbol{r}) = \begin{cases} \Gamma_0, & \boldsymbol{r} \in \Omega_{\mathrm{trap}}, \\ 0, & \boldsymbol{r} \notin \Omega_{\mathrm{trap}}. \end{cases} \tag{6.1}$$

Then the trap contribution is

$$\left.\frac{\partial n_{\mathrm{qp}}}{\partial t}\right|_{\mathrm{trap}} = -\Gamma_{\mathrm{trap}}(\boldsymbol{r}) n_{\mathrm{qp}}. \tag{6.2}$$

This says that the local density decays exponentially in a trap region if diffusion and sources are absent.

A more realistic model would make Γ_{trap} depend on interface transparency, inverse proximity effects, material gap mismatch, and phonon bottleneck physics. Those details belong to a later extension. The first implementation uses Γ_{trap} as an effective design parameter.

7 Thermal material coefficients below T_c

Superconductivity changes thermal coefficients because electronic excitations are gapped. A reduced implementation can use the following modeling hierarchy:

1. Use measured or literature values when available.
2. If unavailable, use normal-state properties above T_c.
3. Below T_c, suppress electronic contributions using an exponential factor controlled by $\Delta/k_{\mathrm{B}}T$.
4. Keep phonon thermal transport in the substrate distinct from electronic/quasiparticle transport in the metal film.

For example, an engineering-level suppression model for an electronic thermal conductivity contribution is

$$k_{e,s}(T) \approx k_{e,n}(T)\,\exp\left[-\frac{\Delta(T)}{k_{\mathrm{B}}T}\right], \tag{7.1}$$

where $k_{e,n}$ is the normal-state electronic contribution. This is not a substitute for detailed superconducting transport theory; it is a reduced model for first geometry sweeps.

8 Coupling heat and quasiparticles

A thermal model and a quasiparticle model can be coupled in several ways. The simplest one-way coupling is

$$T(\boldsymbol{r}, t) \longrightarrow \Delta(T), \quad n_{\mathrm{qp}}^{\mathrm{eq}}(T), \quad D_{\mathrm{qp}}(T), \quad R(T). \tag{8.1}$$

The quasiparticle equation then evolves with coefficients determined by the temperature field.

A stronger two-way coupling adds an energy source/sink to the thermal equation:

$$C\frac{\partial T}{\partial t} = \nabla \cdot (k\nabla T) + Q_{\mathrm{ext}} + Q_{\mathrm{rec}} - Q_{\mathrm{escape}}, \tag{8.2}$$

where Q_{rec} is energy released by recombination and Q_{escape} is energy removed by phonon escape or a heat sink. This is deferred until after the reduced transport model is stable.

9 Boundary and interface modeling

For the first FEM implementation, use no-flux boundary conditions for quasiparticles on insulating boundaries:

$$-D_{\mathrm{qp}}\nabla n_{\mathrm{qp}} \cdot \boldsymbol{n} = 0. \tag{9.1}$$

At trap surfaces or trap volumes, either a volume sink $\Gamma_{\mathrm{trap}} n_{\mathrm{qp}}$ or a Robin boundary condition can be used. A Robin trap boundary has the form

$$-D_{\mathrm{qp}}\nabla n_{\mathrm{qp}} \cdot \boldsymbol{n} = s_{\mathrm{trap}} n_{\mathrm{qp}}, \tag{9.2}$$

where s_{trap} is an interface capture velocity. The volume-sink model is easier to implement in the first 3D mesh, so that is the default.

10 Recommended first model

The first superconducting-temperature model should solve

$$\frac{\partial n_{\mathrm{qp}}}{\partial t} = \nabla \cdot (D_{\mathrm{qp}} \nabla n_{\mathrm{qp}}) - R n_{\mathrm{qp}}^2 - \Gamma_{\mathrm{trap}}(\boldsymbol{r}) n_{\mathrm{qp}} + S_{\mathrm{qp}}(\boldsymbol{r}, t), \tag{10.1}$$

with constant D_{qp} and R inside the superconducting metal, a localized source near the junction or pad, and trap-rate regions defined by geometry tags. Once this works, temperature dependence can be activated through helper functions.

11 References

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